

Unsupervised and Supervised Learning Approaches for Stroke Prediction

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1 Introduction and Research Objectives

The primary research goals for this project are divided into two distinct phases. For the unsupervised phase, the goal is simply to discover natural, underlying structures within the patient data. We want to see if the population can be structurally partitioned into distinct risk profiles without giving the algorithm any prior knowledge whatsoever about stroke outcomes. In the supervised phase, The primary aim is trying to build a predictive model that would be used as a first screening tool to lower the number of undiagnosed stroke cases. Alongside this, the goal is to highlight the algorithmic failure modes caused by extreme class imbalance. Specifically, I want to show that training predictors on raw, unsampled data forces them to be lazy. To mathematically minimize error, they will just default to continuously predicting the majority class, meaning they predict "no stroke" all the time. This creates a dangerous illusion of high overall accuracy, but from a clinical standpoint, the model remains completely blind to the actual stroke cases.

2 Dataset

Finding a dataset that actually challenges the algorithms rather than just acting as a practice dataset is surprisingly difficult. I deliberately avoided the overused datasets typically found in tutorials. Instead, I selected a clinical dataset focusing on stroke prediction. It offers a highly realistic, messy problem typical of health datasets, severe class imbalance and an blend of continuous metabolic markers and categorical demographic data.

The dataset that was used for both parts of the project was obtained from [Kaggle](#) and contains 5,109 records (5,110 before cleaning) and 12 variables, including both categorical and numerical features. The most important characteristic of this dataset is its strong class imbalance: only around 5% of the patients in the sample suffered a stroke. This reflects a realistic medical setting, where stroke cases are relatively rare compared to non-stroke cases, but remain clinically serious and important to detect.

Variable	Type	Description	Role
Gender	Categorical	Patient gender	Feature
Age	Continuous	Patient age in years	Feature
Hypertension	Binary	Presence of hypertension	Feature
Heart Disease	Binary	Presence of heart disease	Feature
Ever Married	Categorical	Marital status	Feature
Work Type	Categorical	Employment type	Feature
Residence Type	Categorical	Urban or rural residence	Feature
Average Glucose Level	Continuous	Blood glucose level	Feature
BMI	Continuous	Body mass index	Feature
Smoking Status	Categorical	Smoking behaviour	Feature
Stroke	Binary	Stroke outcome	Target

Table 1: Dataset variables, types, descriptions, and roles

3 Unsupervised Learning: Dimensionality Reduction

Applying standard Principal Component Analysis (PCA) to this dataset would be mathematically flawed because PCA is strictly restricted to continuous variables. Throwing out categorical factors like marital status or work type would mean discarding vital clinical variance. Therefore, Factor Analysis of Mixed Data (FAMD) was implemented to handle the heterogeneous data.

Before running the FAMD algorithm, a specific encoding strategy was required. Binary clinical indicators (like heart disease) were recoded into mathematically unique strings (e.g., "Yes_HD", "No_HD"). This wasn't just a technical workaround to stop the FactoMineR package from crashing due to row-name duplication errors; it preserved the discrete, non-ordinal nature of these categorical variables in the projection space.

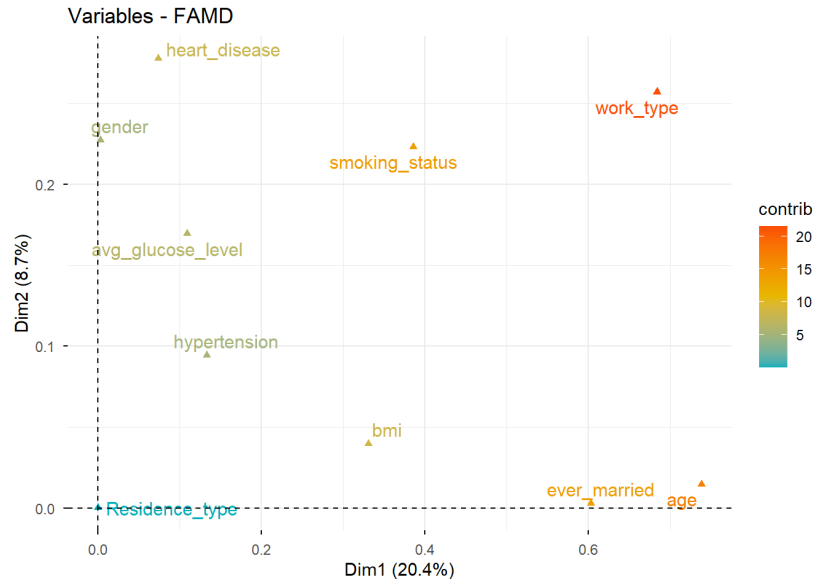


Figure 1: Variable Contributions in the First Two FAMD Dimensions

The resulting variables plot highlighted a clear structural divide: Dimension 1 is heavily loaded with aging and demographic factors, while Dimension 2 is driven by cardiovascular and metabolic indicators. Plotting the individuals—and then retroactively overlaying their

true stroke status revealed that stroke cases cluster in the positive regions of both dimensions.

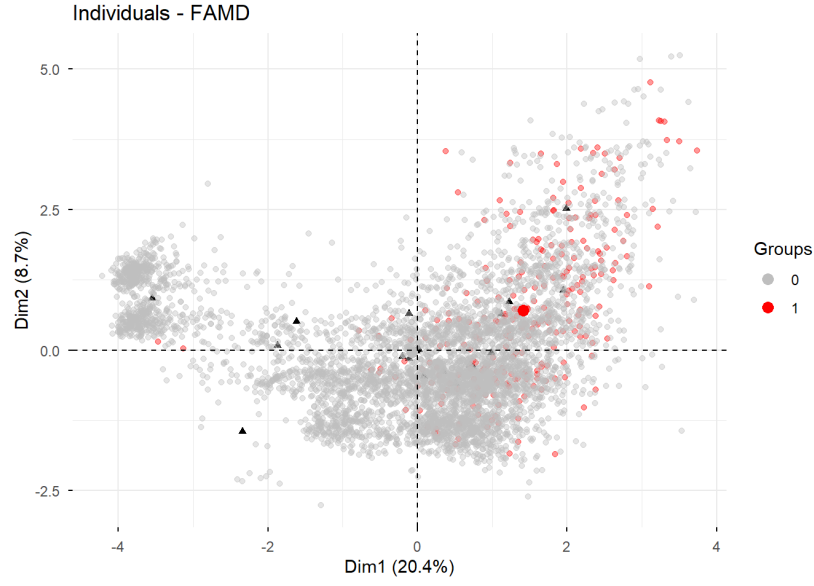


Figure 2: Projection of patients onto the FAMD dimensions by stroke status

3.1 Unsupervised Learning: Clustering

Before beginning with clustering I decided to use the Kaiser criterion to evaluate which dimensions to retain, and as the scree plot demonstrates, all the dimensions with an eigenvalue $\lambda > 1.0$ were kept, meaning the top 5 principal dimensions.

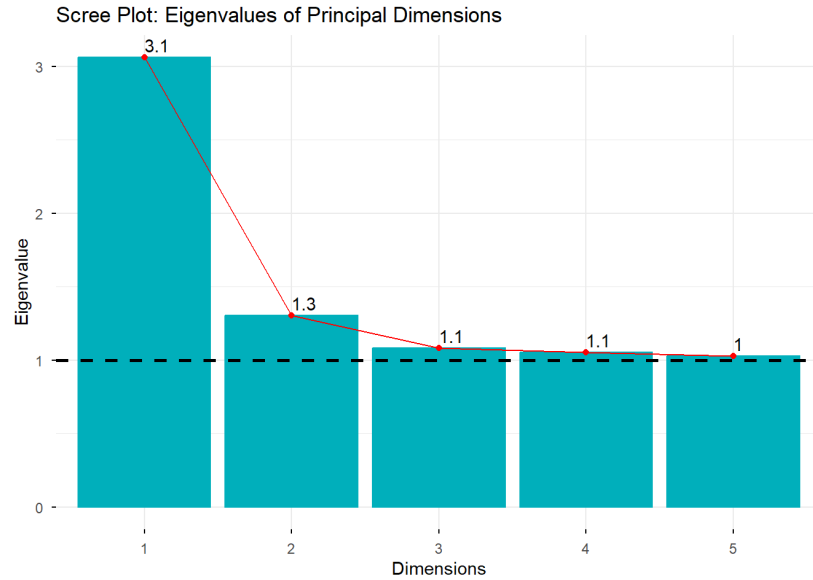


Figure 3: Scree Plot of FAMD Eigenvalues

Since standard K-Means clustering relies on Euclidean distance, applying it directly to raw mixed data is invalid. To bypass this, K-Means was applied to the numerical coordinates extracted from the FAMD dimensions. Evaluating the Total Within Sum of Square (WSS)

using the elbow method showed a steep decline that flattened significantly after $k = 3$.

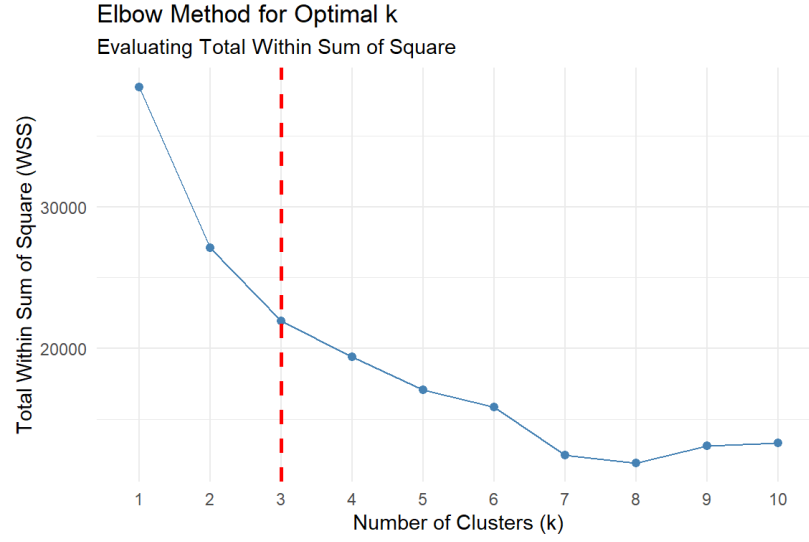


Figure 4: Elbow Method for Selecting the Optimal Number of K-Means Clusters

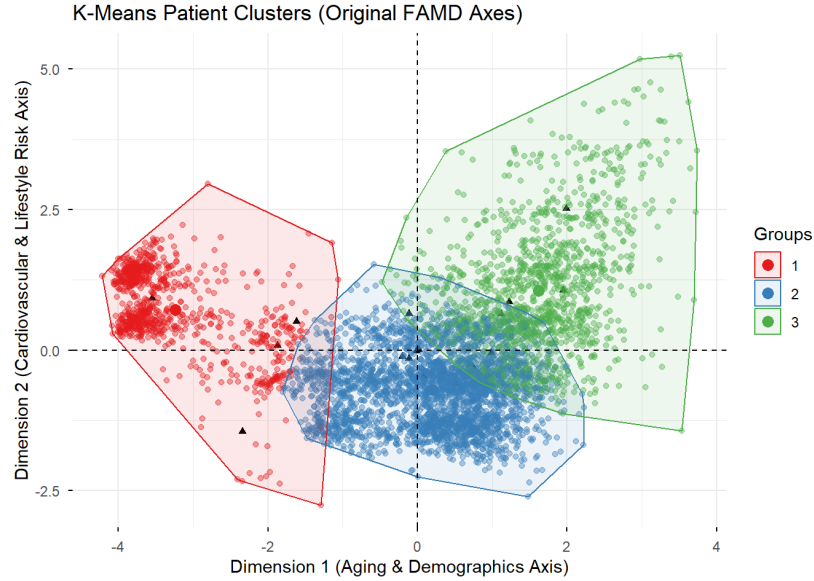


Figure 5: K-Means clustering results projected onto the FAMD dimensions

The K-Means clustering successfully showed that the patients naturally partition in 3 distinct clinical profiles:

- **Cluster 1:** This group mainly represents younger individuals with generally favourable health characteristics and a very low observed incidence of stroke.
- **Cluster 2:** This cluster captures a more intermediate profile, mostly composed of middle-aged individuals with moderate metabolic indicators and a risk level between the other two groups.
- **Cluster 3:** This group represents the clearest high-risk profile. It is characterized by

older age, higher average glucose levels, and a greater prevalence of hypertension. As expected, this cluster contained the largest concentration of actual stroke cases.



Figure 6: Stroke rate by cluster

Cluster	Total Patients	Mean Age	Mean Glucose	Stroke Rate (%)
1	850	9	93.9	0.24
2	2903	43	94.6	2.86
3	1356	65	138.4	12.09

Table 2: Statistics of the three clusters

4 Supervised Learning: Handling Imbalance & Project Setup

Transitioning to predictive classification, the dataset’s severe class imbalance needs to be addressed before any model can be applied. The stroke cases represented a tiny fraction of the observations. If left untreated, a model would simply learn to predict ”No Stroke” every single time, achieving high accuracy while being clinically useless.

To rectify this, I applied SMOTENC (Synthetic Minority Over-sampling Technique for Nominal and Continuous). SMOTENC was explicitly chosen over standard SMOTE because our feature space contains categorical predictors, which standard Euclidean-based SMOTE cannot interpolate correctly. The minority class was oversampled until it reached 25% of the majority class size within the training partition only.

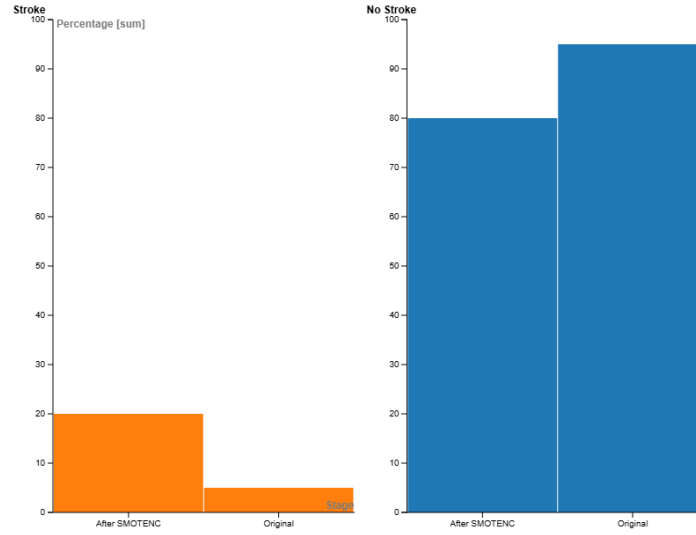


Figure 7: Class distribution in the training set before and after SMOTENC resampling

4.1 Supervised Learning: Model Selection & Hyper-Parameter Tuning

Three models were evaluated: Logistic Regression (as a highly interpretable baseline), Random Forest, and XGBoost. The tree-based models were included to capture potential non-linear relationships that a linear model might miss. Cross-validation was performed using 5 stratified folds.

Initially, tuning the Random Forest and XGBoost models for ROC-AUC yielded excellent overall accuracy but really low sensitivity (8.8% and 3.0%, respectively). They were functionally blind to actual stroke patients.

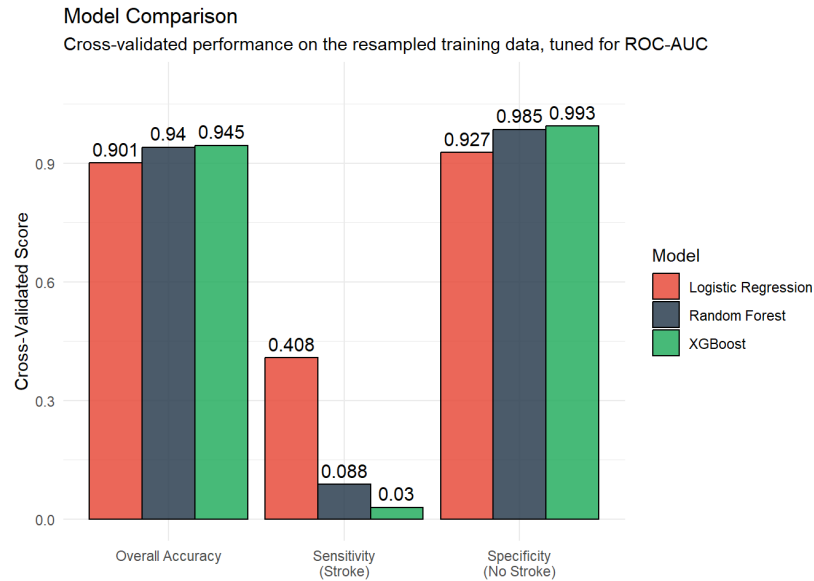


Figure 8: Cross-validated comparison of model performance after tuning for ROC-AUC

However, when the grid search was explicitly re-configured to optimize for sensitivity, performance shifted dramatically. The sensitivity for Random Forest jumped to 22.9%, and

XGBoost reached 26.0%.

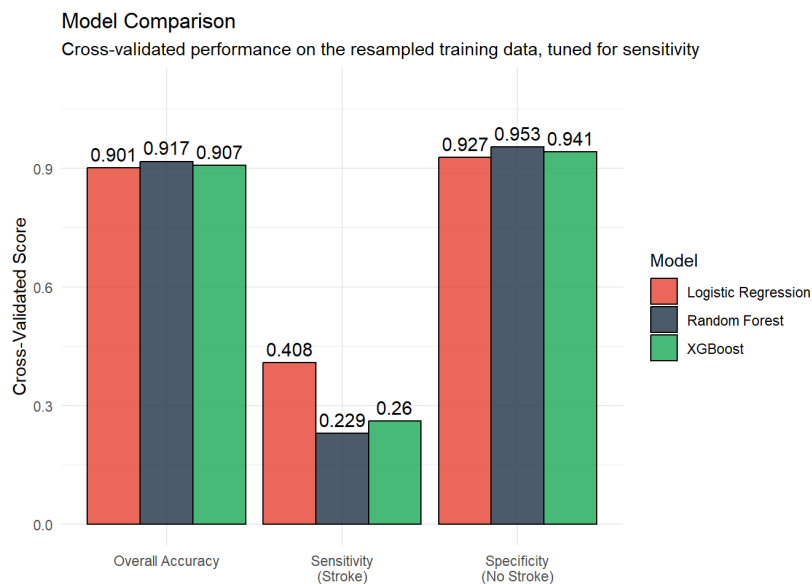


Figure 9: Cross-validated comparison of model performance after tuning for sensitivity

Yet, the Logistic Regression model still outperformed them, achieving a 40.8% sensitivity rate with only a marginal drop in overall accuracy. A comparative test against an unsampled Logistic Regression model confirmed that without SMOTENC, the baseline model was merely guessing the majority class. This proved that the resampling step was necessary.

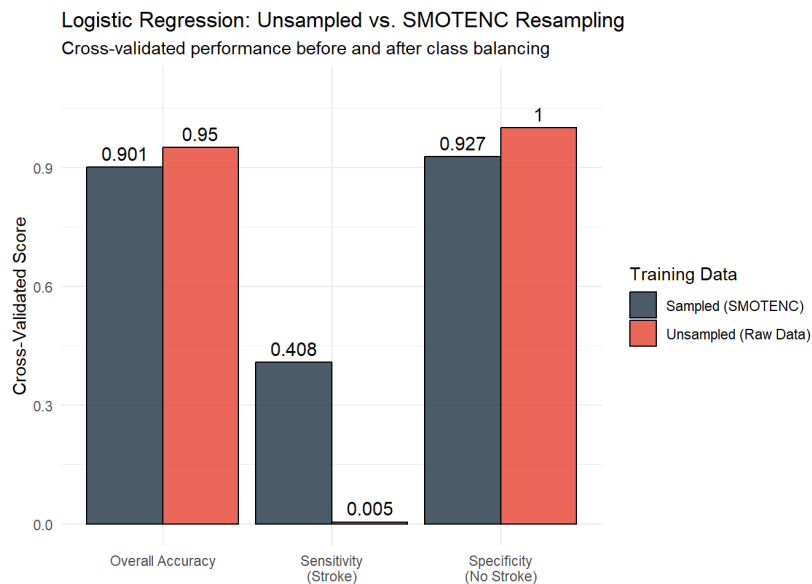


Figure 10: Comparison of Logistic Regression performance with and without SMOTENC resampling

4.2 Supervised Learning: Decision Threshold Adjustment

The default probability threshold for binary classification is typically $p = 0.50$. In a medical screening context, this can be adjusted depending on the goal. In the bar chart below, it can

be shown that lowering this threshold can yield better sensitivity but at the cost of a slight decrease in overall accuracy and an higher number of false-positives.



Figure 11: Logistic Regression performance across different probability thresholds

4.3 Supervised Learning: Evaluation on Test Set & Conclusions

By analyzing the results across different decision boundaries, I determined that lowering the threshold to $p = 0.15$ significantly enhanced detection capabilities. Applying this final threshold to the completely unseen test set resulted in catching 35 out of 44 actual stroke cases (a final sensitivity of roughly 79.5%). Yes, this generated 324 false positives. However, from a health economics perspective, this may be the trade off a hospital wants to make. The cost of running a few secondary, preventative tests on healthy individuals is minimal compared to the high medical and financial burden of missing a stroke diagnosis. The tuned Logistic Regression model, therefore, proves to be an effective, cautious first-line screening tool.

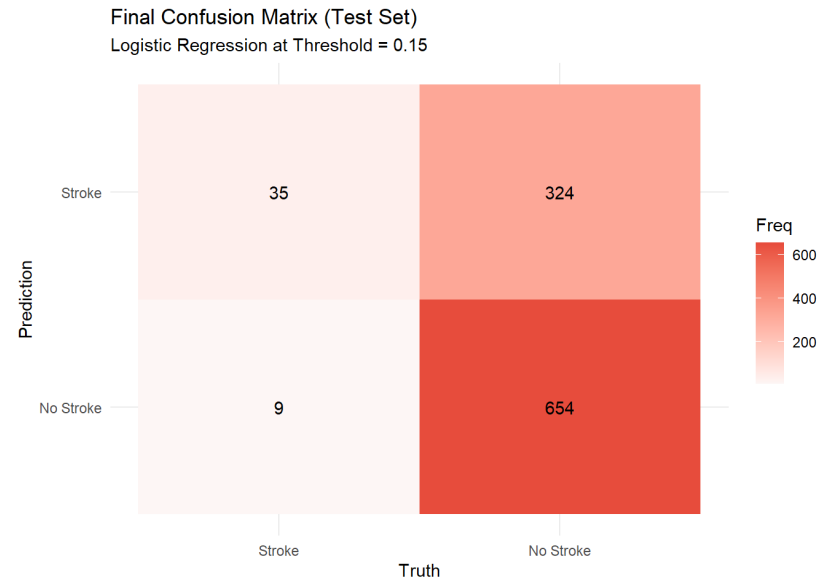


Figure 12: Final test-set confusion matrix